

BLOOD TESTS

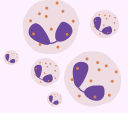
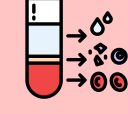
INTRODUCTION

- Blood tests are essential tools in nursing practice, providing vital information about a patient's overall health and guiding clinical decisions. They help detect, monitor, and evaluate medical conditions, ensuring timely and accurate care



COMPLETE BLOOD COUNT (CBC)

A CBC evaluates overall health, detects infections, anemia, clotting problems, and blood cancers

COMPONENT	NORMAL	HIGH	LOW	NURSING IMPLICATIONS
White Blood Cells (WBCs) 	4,500–11,000 /mm ³	Leukocytosis: - Infection - Inflammation - leukemia - Steroid use	Leukopenia: - Chemotherapy - Radiation - Autoimmune disease - Bone marrow Suppression	1. Neutropenic precautions if ANC <1,000: hand hygiene, avoid raw foods, limit visitors 2. Fever in neutropenic patient = medical emergency
Red Blood Cells (RBCs) 	Male: 4.7–6.1 million Female: 4.2–5.4 million	Polycythemia: - COPD - Dehydration - High altitude	Anemia: - Blood loss, - Iron deficiency - Chronic kidney disease	1. Low RBC count parallels low hemoglobin 2. Always assess both together when evaluating anemia
Hemoglobin (Hgb) 	Male: 13–18 g/dL Female: 12–16 g/dL.	- Anemia - Hemorrhage	Critical: <7 g/dL → transfusion	Tip: Each 1g/dL drop in Hgb reduces oxygen-carrying capacity, so monitor for hypoxia signs even if vitals look stable
Hematocrit (Hct) 	Male: 42–52% Female: 37–47%.	- Dehydration - Polycythemia	Anemia Fluid overload	Tip: Hematocrit ≈ 3x Hgb
Platelets (Plt) 	150,000–400,000 /mm ³	Thrombocytopenia (<50,000): bleeding risk	<20,000: spontaneous bleeding → critical	No IM injections, soft toothbrush, no aspirin

BLOOD TESTS



COAGULATION STUDIES

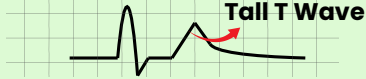
Used to monitor bleeding/clotting risk

Tip: Always check bleeding risk before invasive procedures if the patient is on anticoagulants

COMPONENT	KEY POINTS
PT (11–14 sec)	Prolonged with warfarin.
INR (0.9–1.2 normal)	2–3 therapeutic on warfarin 4 = critical bleeding risk
aPTT (25–35 sec normal)	60–80 therapeutic on heparin 80 = high bleeding risk
D-Dimer (<0.5)	Elevated = clot breakdown → PE, DVT, DIC



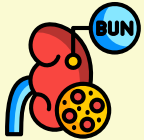
BASIC METABOLIC PANEL (BMP) ELECTROLYTES

ELECTROLYTE	HIGH	LOW
Sodium (135–145 mEq/L) 	- Dehydration - Cushing's syndrome - Diabetes Insipidus	- SIADH - CHF - Fluid overload
Potassium (3.5–5.0 mEq/L) 	- Renal failure, acidosis - K-sparing diuretics → peaked T waves 	- Diuretics → flat T waves, U wave - Vomiting - Diarrhea 
Calcium (8.5–10.5 mg/dL) 	- Bone pain - Kidney stones - Confusion	- Tetany - Seizures - Chvostek's & Trousseau's signs
Magnesium (1.5–2.5 mEq/L) 	↓ Reflexes - Respiratory depression	- Tremors - Seizures ↑ Reflexes

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BASIC METABOLIC PANEL (BMP)

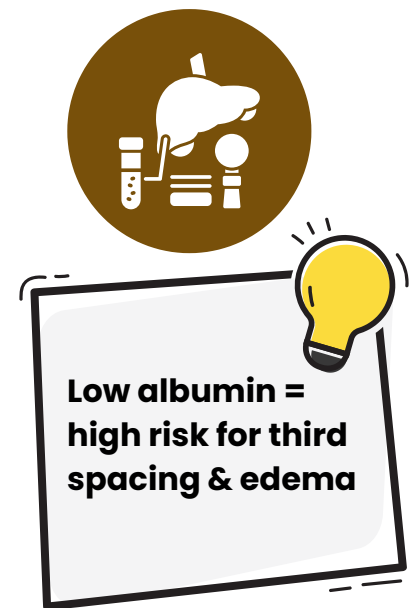
KIDNEY FUNCTION TESTS

ELECTROLYTE	KEY POINTS
BUN (10–20 mg/dL)	High: dehydration, renal impairment, GI bleed 
Creatinine (0.6–1.2 mg/dL)	- More specific for renal function - Elevated = kidney damage Note: Creatinine >2 = critical renal impairment 

LIVER FUNCTION TESTS (LFTs)

Evaluate liver health

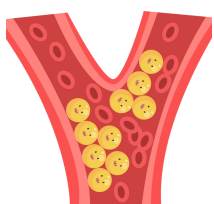
ALT (10–40)	Elevated = hepatitis, cirrhosis.
AST (10–40)	Elevated in liver, cardiac, and muscle injury
ALP (44–147)	Biliary obstruction, bone disease.
Bilirubin (0.3–1.2)	High = jaundice, hemolysis.
Albumin (3.5–5.0)	Low = ascites, edema (↓ oncotic pressure)



LIPID PANEL

Assesses risk for cardiovascular disease.

 **Tip:** High LDL + low HDL = high atherosclerosis risk



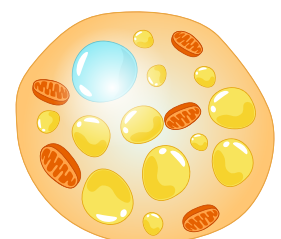
Total Cholesterol
<200 mg/dL



HDL ("good")
>40 (men), >50 (women)



LDL ("bad")
<100 mg/dL



Triglycerides
<150 mg/dL

BLOOD TESTS

CARDIAC ENZYMES

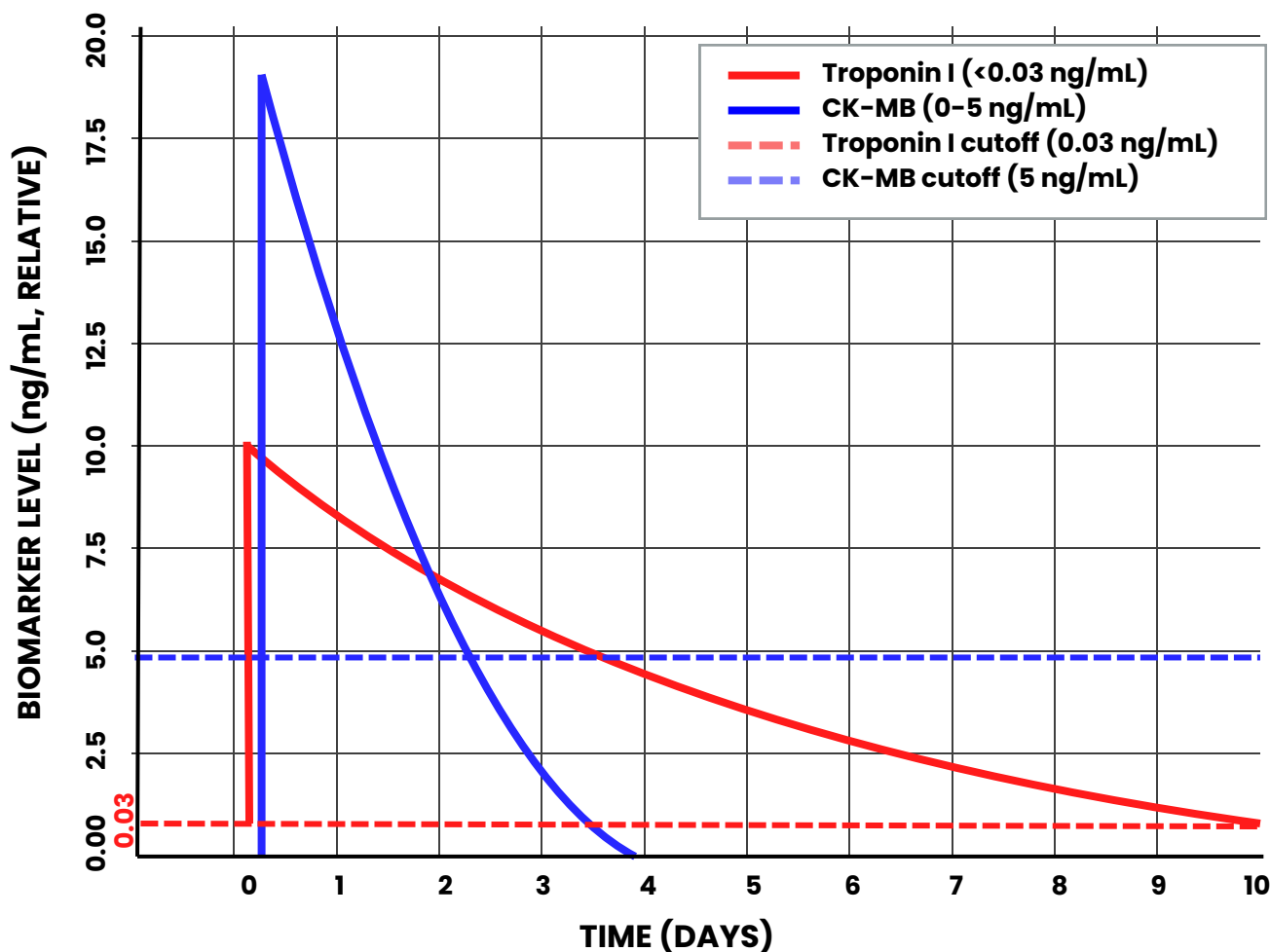
Detect myocardial infarction (MI)



Tip: Troponin = gold standard for MI diagnosis

- **Troponin I (<0.03 ng/mL):** Most specific. Rises in 3–6 hrs, stays 7–10 days
- **CK-MB (0–5 ng/mL):** Rises in 4–6 hrs, normal in 2–3 days
- **BNP (<100 pg/mL):** $\uparrow$ = Heart failure

BIOMARKER KINETICS AFTER MYOCARDIAL INFARCTION



DIABETES-RELATED TESTS

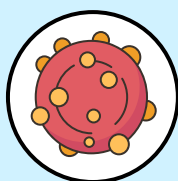


Tip: HbA1c shows 3-month average blood sugar



Fasting Glucose

- 70–100 mg/dL
- 126 = diabetes



HbA1c

- $<5.7\%$ normal
- $5.7\text{--}6.4\%$ prediabetes
- $\geq 6.5\%$ = diabetes



Random Glucose

- >200 mg/dL with symptoms = diabetes

BLOOD TESTS

ARTERIAL BLOOD GASSES (ABGs)

VALUE	NORMAL	MEANING IF ↑	MEANING IF ↓
pH 	7.35–7.45	Alkalosis	Acidosis
PaCO₂ 	35–45	Respiratory acidosis	Respiratory alkalosis
HCO₃⁻ 	22–26	Metabolic alkalosis	Metabolic acidosis
PaO₂ 	80–100	Hyperoxygenation	Hypoxemia
SaO₂ 	95–100%	–	Hypoxia

QUICK INTERPRETATION OF ABGs

- pH ↓ & CO₂ ↑ = Respiratory Acidosis
- pH ↑ & CO₂ ↓ = Respiratory Alkalosis
- pH ↓ & HCO₃ ↓ = Metabolic Acidosis
- pH ↑ & HCO₃ ↑ = Metabolic Alkalosis

Tip: Use ROME (Respiratory Opposite, Metabolic Equal)

POINTS TO REMEMBER

1 Potassium & Calcium = cardiac priority

2 Hgb <7 g/dL → transfuse.

3 INR >4 = bleeding risk

4 Troponin = MI confirmation

5 Albumin ↓ = ascites, edema.

6 Neutropenic patients = strict infection precautions

BLOOD TESTS



URINALYSIS

Urinalysis evaluates kidney function, hydration status, and presence of infection or systemic disease

NORMAL URINALYSIS VALUES

Component	Normal Range/Appearance	Nursing Notes
Color	Pale yellow to amber	Dark = dehydration, tea-colored = liver disease
Clarity	Clear to slightly cloudy	Cloudy = infection, crystals, protein
Odor	Faint/aromatic	Fruity = DKA; foul = infection
pH	4.5–8.0	Acidic = metabolic/resp acidosis; Alkaline = UTI, diet high in citrus/vegetables
Specific Gravity	1.005–1.030	Low = dilute (DI, excess fluids); High = concentrated (dehydration, SIADH)
Protein	Negative	Proteinuria = kidney disease, preeclampsia
Glucose	Negative	Glycosuria = uncontrolled diabetes, pregnancy
Ketones	Negative	Positive = DKA, starvation, fasting
Blood (Hematuria)	Negative	Positive = infection, stones, trauma, cancer
Leukocyte Esterase	Negative	Positive = WBCs → UTI
Nitrites	Negative	Positive = bacterial infection (E. coli common)
Bilirubin	Negative	Positive = liver disease, bile duct obstruction
Urobilinogen	0.1–1.0 mg/dL	↑ in hemolysis, liver disease
Microscopic Analysis	Occasional RBCs, WBCs few epithelial cells	↑ RBCs = stones, trauma; ↑ WBCs = UTI

BLOOD TESTS



ABNORMAL URINALYSIS REPORT

PROTEINURIA	- Kidney disease (nephrotic syndrome, glomerulonephritis) - Preeclampsia
GLYCOSURIA	- Diabetes mellitus - Pregnancy
KETONURIA	- Diabetic ketoacidosis - Starvation & low-carb diet
HEMATURIA	- Stones - Trauma - Infection - Tumors
LEUKOCYTE ESTERASE + NITRITES	Urinary tract infection (UTI)
BILIRUBIN	Hepatitis, biliary obstruction.
UROBILINOGEN ↑	Hemolytic anemia, cirrhosis.

NURSING IMPLICATIONS

1. Specimen Collection

- Use clean-catch **midstream** for culture & sensitivity
- Send to lab **within 1 hour** or refrigerate
- Label correctly (patient name, date, time)
- Educate patient: **avoid contamination** with stool/menstrual blood

2. Patient Education

- Morning urine = **best sample** (most concentrated)
- Encourage hydration before collection unless contraindicated

3. Interpreting Results

- Look at trends (persistent proteinuria = chronic disease)
- Always correlate with patient symptoms (e.g., UTI signs + UA findings).



SPECIAL TESTS IN URINALYSIS

- **Urine Culture & Sensitivity (C&S):** Identifies bacteria + antibiotic sensitivity
- **24-Hour Urine Collection:** Measures creatinine clearance, protein. → Requires discarding first specimen, then collecting all urine for 24 hrs
- **Drug Screen:** Detects substances (amphetamines, opioids, etc.)

POINTS TO REMEMBER

DON'T FORGET

1. **Specific gravity high** = dehydration (concentrated urine)
2. **Protein + HTN in pregnant woman** = preeclampsia
3. **Glucose + ketones** = diabetic ketoacidosis
4. **Leukocyte esterase + nitrites** = UTI
5. **Bilirubin + urobilinogen** = liver disease

TIP

Always discard the first urine sample in a 24-hour collection, then keep all others.
Missing a sample = restart